

Python Practice Handout

Booleans and if statements

1. What is a Boolean

You already know int, float and string. A Boolean is the fourth type. It represents a yes or no situation, and it has only two possible values: True or False.

A good example is whether a light is switched on or off.

```
light_is_on = True
```

2. No quotes, and a capital letter

True and False must start with a capital letter and must not have quotes around them. If you put quotes around it, you no longer have a Boolean. You have a string containing the word.

A Boolean:

```
light_is_on = False
```

A string, not a Boolean:

```
light_is_on = "False"
```

3. The if statement

A Boolean becomes useful with an if statement. Write if, then the variable, then a colon. The line underneath runs only when the Boolean is True.

```
light_is_on = True

if light_is_on:
    print("The light is on")
```

Output:

```
The light is on
```

Now change it to False and run again. Nothing is printed at all.

```
light_is_on = False

if light_is_on:
    print("The light is on")
```

Output:

(nothing)

4. The tab matters

The line under an if statement is moved across by pressing Tab. Every tabbed line belongs to the if and only runs when it is True. A line that is not tabbed over is outside the if, so it always runs.

```
light_is_on = False

if light_is_on:
    print("The light is on")
    print("Hello")
print("Goodbye")
```

Output:

```
Goodbye
```

The first two prints are tabbed over so they are skipped. The last one is not tabbed, so it runs either way.

5. Try it yourself

Make a Boolean variable for whether the computer you are using is switched on. Then print a message if it is. Try this on your own before looking below.

```
computer_is_on = True

if computer_is_on:
    print("The computer is on")
```

6. else

Often you want something to happen when the Boolean is False as well. Add else with a colon underneath. Make sure else is not tabbed over, but the line below it is.

```
light_is_on = False

if light_is_on:
    print("The light is on")
else:
    print("We are in the dark")
```

Output:

```
We are in the dark
```

Change the variable to True and the first message prints instead. One of the two always runs.

7. Comparisons make Booleans

You do not need a Boolean variable. Comparing two things produces True or False on the spot, and an if statement can use that directly.

```
weight = 190.4

if weight < 195:
    print("You're under 195")
```

Output:

```
You're under 195
```

8. The comparison signs

There are six comparisons. Printing one on its own shows that it really does produce a Boolean.

```
weight = 190.4

print(weight < 195)
print(weight > 195)
print(weight <= 195)
print(weight >= 195)
print(weight == 195)
print(weight != 195)
```

Output:

```
True
False
True
False
False
True
```

Less than and greater than can also include equal to. Two equal signs asks whether the two things are the same. An exclamation mark with an equal sign asks whether they are different.

9. Try it yourself

Make a variable called age. If the person is 18 or older, print that they are an adult, otherwise print that they are a child. Try this on your own before looking below.

```
age = 32

if age >= 18:
    print("Adult")
else:
    print("Child")
```

Output:

```
Adult
```

Try it again with 18, and then with 17, and check that you get what you expect.